

Task 73: Longest Nice Substring

PROBLEM DESCRIPTION

A string *s* is nice if, for every letter of the alphabet that *s* contains, it appears both in uppercase and lowercase. For example, 'abABB' is nice because 'A' and 'a' appear, and 'B' and 'b' appear. However, 'abA' is not because 'b' appears, but 'B' does not. Given a string *s*, return the longest substring of *s* that is nice.

JAVA SOLUTION

```
import java.util.HashSet;
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class Solution {
    public String longestNiceSubstring(String s) {
        if (s.length() < 2) return "";

        Set<Character> set = new HashSet<>();
        for (char c : s.toCharArray()) {
            set.add(c);
        }

        for (int i = 0; i < s.length(); i++) {
            char c = s.charAt(i);
            if (set.contains(Character.toLowerCase(c)) &&
                set.contains(Character.toUpperCase(c))) {
                continue;
            }

            // Divide and Conquer
            String left = longestNiceSubstring(s.substring(0, i));
            String right = longestNiceSubstring(s.substring(i + 1));

            return left.length() >= right.length() ? left : right;
        }

        return s;
    }
}
```